Cristiano Ronaldo's Champions League Legacy: Exploratory Data Analysis

CSE 163 | Project Part 2 | November 12, 2026 Authors: Rodrigo Li and Larry Zhang Public team GitHub: https://github.com/yiranli7-code/CSE163-UCL-Data-Analysis

1. Summary of Research Questions

RQ1 — Career contributions: Compare Cristiano Ronaldo with Lionel Messi, Robert Lewandowski, Karim Benzema and Kylian Mbappé using UEFA-listed goals, assists and G+A totals.

RQ2 — Efficiency: Examine goals and goal contributions per appearance only when compatible appearance denominators are available.

RQ3 — Seasonal peaks: Compare four selected Ronaldo seasons with standout seasons from other players using recorded goals.

RQ4 — Match prediction: Use regularized logistic regression and tuned random forests to predict whether a match has at least three total goals from competition stage and each side's pre-match scoring form.

2. Motivation

Career accumulation, exposure-adjusted rates and exceptional seasons express different kinds of football performance. Our goal is a transparent comparison, not a predetermined "greatest player" conclusion. The match-level model is a separate statistical-learning application, not evidence of Ronaldo's individual causal impact on Real Madrid outcomes.

3. Datasets and Provenance

UEFA career snapshot: five selected players × four source attributes (player, goals, assists, appearances). UEFA selected peaks: nine player-season entries × four fields (player, season, goals, source_note). OpenFootball: 13 raw cl.txt season files (2011/12–2023/24) from the uploaded CC0 GitHub archive, converted into 1,619 main-competition matches × eight source fields. The OpenFootball match records do not contain complete player assists or minutes and are not substituted for UEFA's player ranking tables.

4. Method

We parsed scorelines and stage headings, including localized headings, and checked each season's fixture count against its # Matches header. We excluded qualifiers, counted extra-time goals, and omitted shootout conversions. In pandas, we validated dates and fixture uniqueness, calculated total goals and the ≥3-goal indicator, and computed each club's season-to-date scoring and conceding averages using only matches played on earlier dates. The EDA summarizes numerical distributions, missing values, and category frequencies. We used Python to generate eight figures, test the stage association, and evaluate the models.

5. EDA Results: Size, Missingness and Variables

We checked missingness using pandas isna().sum() and summarized categorical distributions using value_counts(). Career table: 5 × 4 raw and 5 × 7 after adding G+A and two rate variables. Each player appears once. Goals, assists, and G+A are complete. Lewandowski and Mbappé lack verified appearance counts (2 of 5, or 40%). We keep both players in the totals and leave them out of the per-appearance rankings until we can verify denominators that use the same competition scope.

Variable	n	Mean	SD	Min	Q1	Med	Q3	Max
Goals	5	108.60	27.43	74.00	90.00	109.00	129.00	141.00
Assists	5	31.80	8.79	22.00	26.00	29.00	40.00	42.00
Appearances	3	167.33	17.90	152.00	157.50	163.00	175.00	187.00
G+A	5	140.40	34.68	100.00	119.00	131.00	169.00	183.00
Goals/app	3	0.71	0.11	0.59	0.67	0.75	0.77	0.79
G+A/app	3	0.93	0.13	0.78	0.88	0.98	1.01	1.04

Peak-season table: 9 × 4, with no missing values. Ronaldo appears four times, Mbappé twice, and Messi, Benzema, and Lewandowski once each. The selected seasons are 2007/08, 2011/12, 2013/14, 2015/16, 2016/17, 2019/20, 2021/22, 2023/24, and 2025/26, with one record each.

Variable	n	Mean	SD	Min	Q1	Med	Q3	Max
Season goals	9	13.33	3.32	8.00	12.00	15.00	15.00	17.00

Match table: 1,619 × 8 parsed fields and 1,619 × 10 after adding total_goals and high_scoring. The stage counts are 1,248 group and 371 knockout fixtures. All 10 columns are complete. Of these fixtures, 208 (12.85%) lack prior-match averages because at least one team has no earlier match that season. We retain all 1,619 fixtures for the match-level EDA and use the 1,411 with complete pre-match features for modeling.

Variable	n	Mean	SD	Min	Q1	Med	Q3	Max
Home goals	1619	1.67	1.44	0.00	1.00	1.00	2.00	8.00
Away goals	1619	1.30	1.29	0.00	0.00	1.00	2.00	8.00
Total goals	1619	2.97	1.73	0.00	2.00	3.00	4.00	12.00
High (0/1)	1619	0.57	0.50	0.00	0.00	1.00	1.00	1.00

Season counts: 125 fixtures per season, except for 2019/20 (119). The home and away team labels identify the two clubs and link each to its earlier matches in the same season. The model uses the resulting scoring and conceding averages plus the competition stage.

5.1 Career Contributions and Efficiency (RQ1–RQ2)

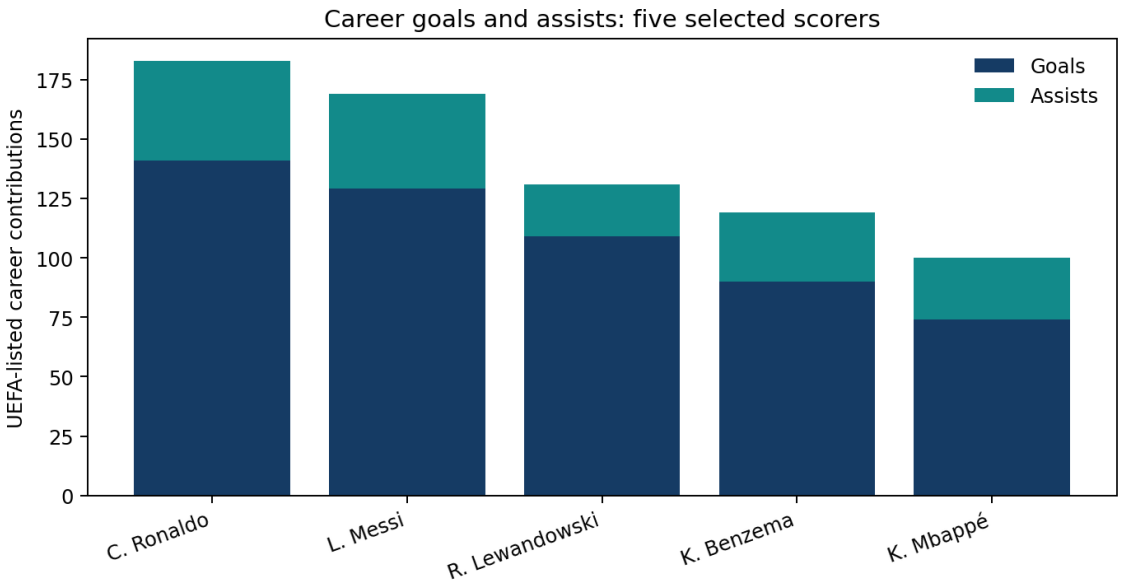


Figure 1. Ronaldo leads the five-player UEFA snapshot in listed career goals and observed goals plus assists.

The stacked bar plot separates goals and assists: Ronaldo has 141 + 42 = 183 listed contributions, compared with Messi's 129 + 40 = 169. The comparison uses one UEFA ranking convention but is not a claim about all competition-stage definitions.

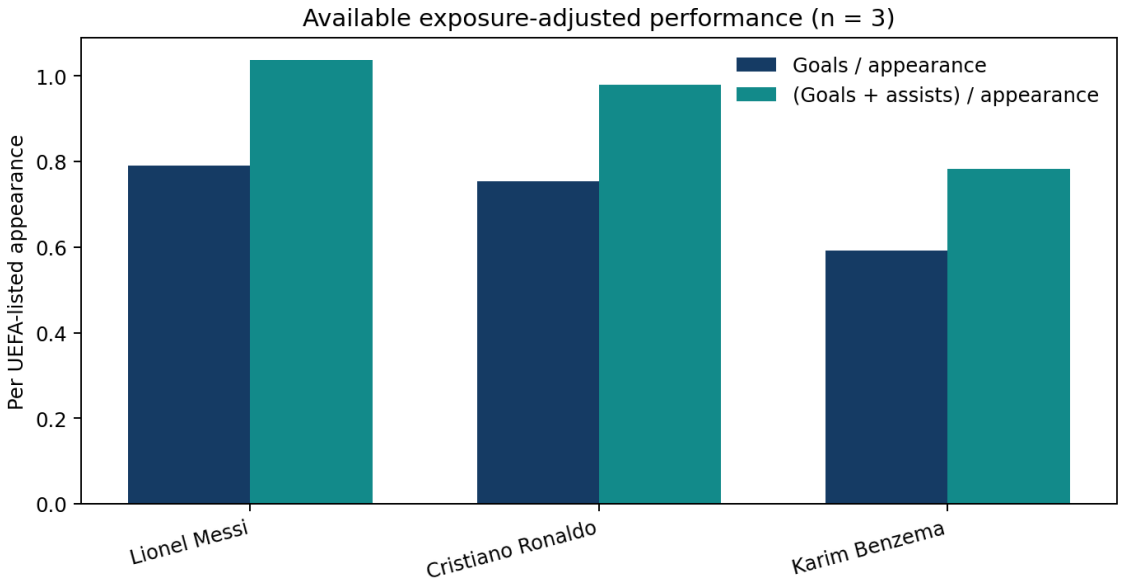


Figure 2. Messi has the highest goals per UEFA-listed appearance among the three players with observed denominators.

The grouped bars show goals and G+A per appearance. Messi leads in goals per appearance (129/163 ≈ 0.79), followed by Ronaldo (141/187 ≈ 0.75) and Benzema (90/152 ≈ 0.59).

5.2 Selected Seasonal Peaks (RQ3)

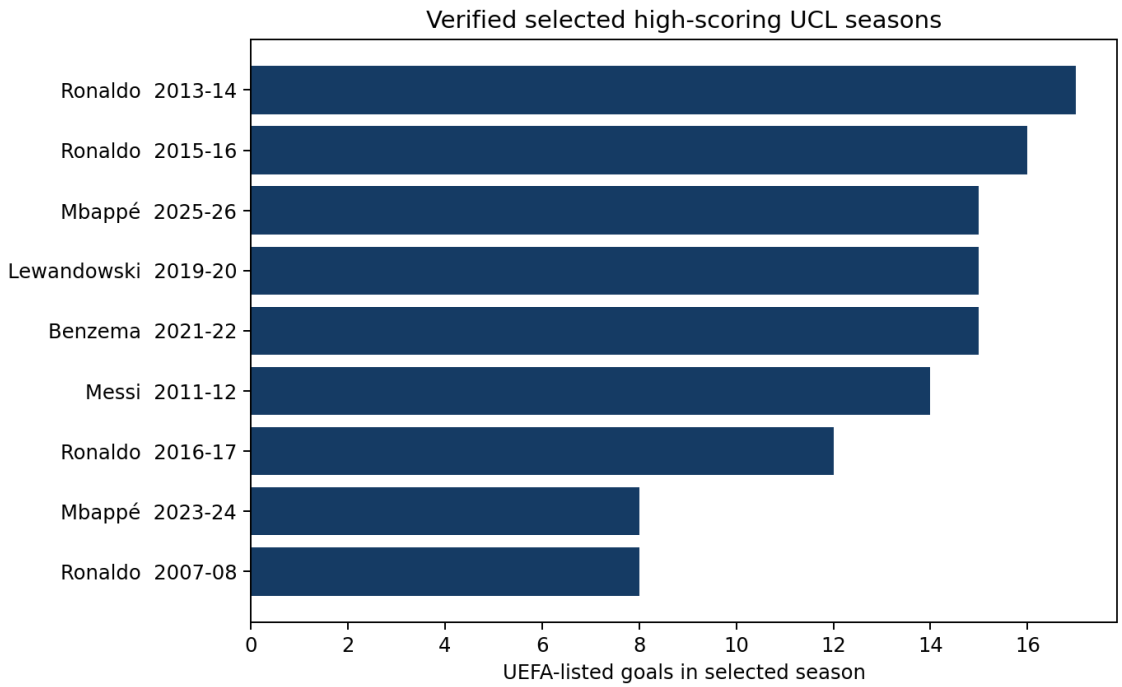


Figure 3. Ronaldo's 2013/14 total of 17 goals is the largest among the nine selected player-seasons.

A horizontal bar plot allows direct comparison of the source season-goal totals, including Ronaldo 2015/16 (16) and Benzema 2021/22 (15). The entries were deliberately selected, so this is not a distribution of all UCL player-seasons.

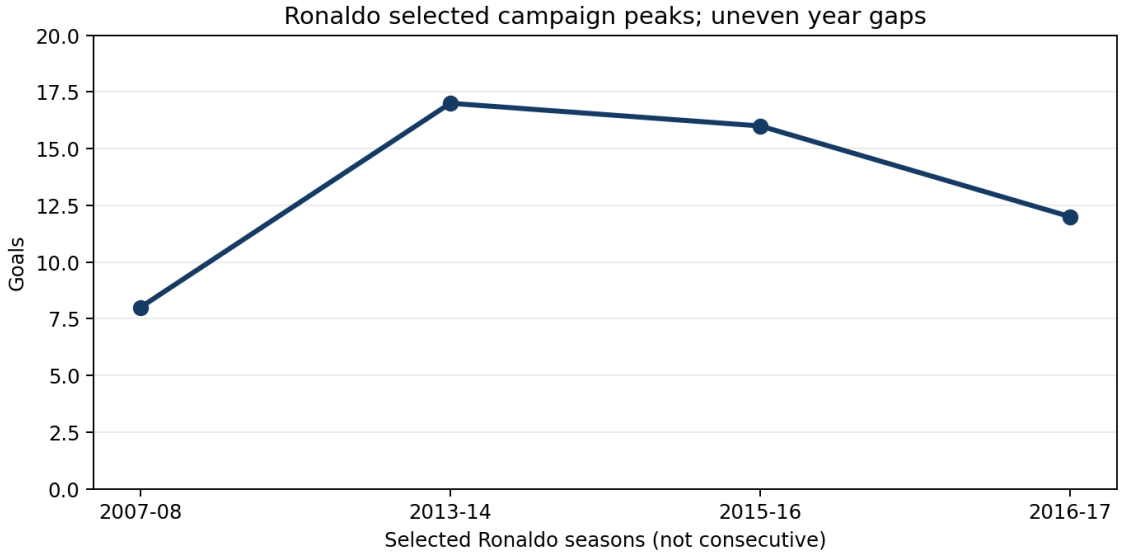


Figure 4. Ronaldo's four highlighted seasons scored 8, 17, 16 and 12 goals, respectively.

The line connects four selected campaigns with uneven gaps between them. It shows the recorded goal totals for those seasons only. Historical assists were unavailable for this comparison.

5.3 Thirteen-Season Match Results (RQ4)

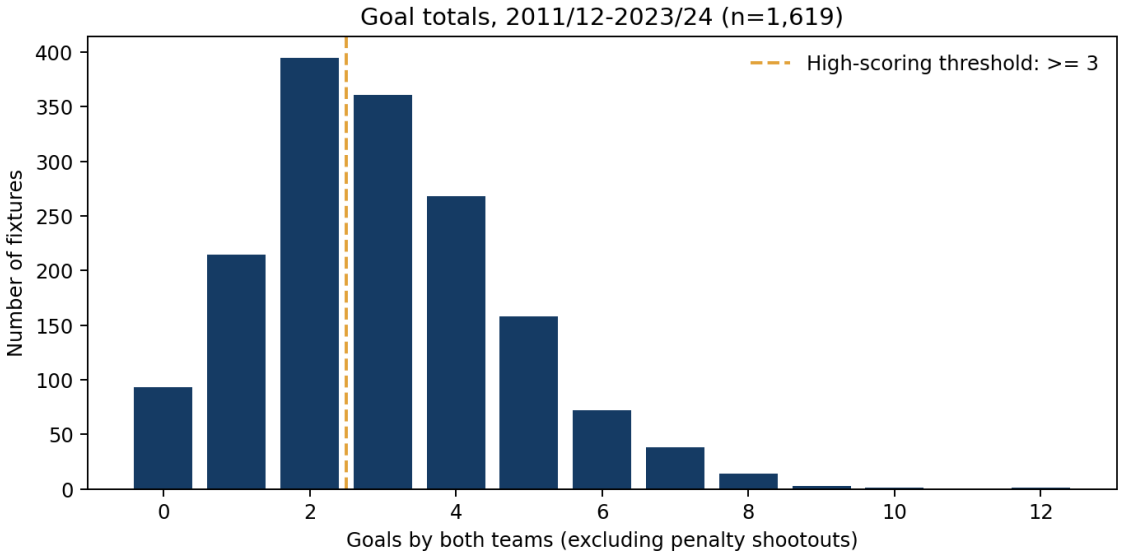


Figure 5. Across 1,619 fixtures, mean scoring was 2.97 goals per match and 916 games (56.6%) reached three goals.

This discrete goal-count plot gives the distribution underlying the binary target; totals range from 0 to 12. The dashed threshold marks three goals, and extra-time goals count but shootout conversions do not.

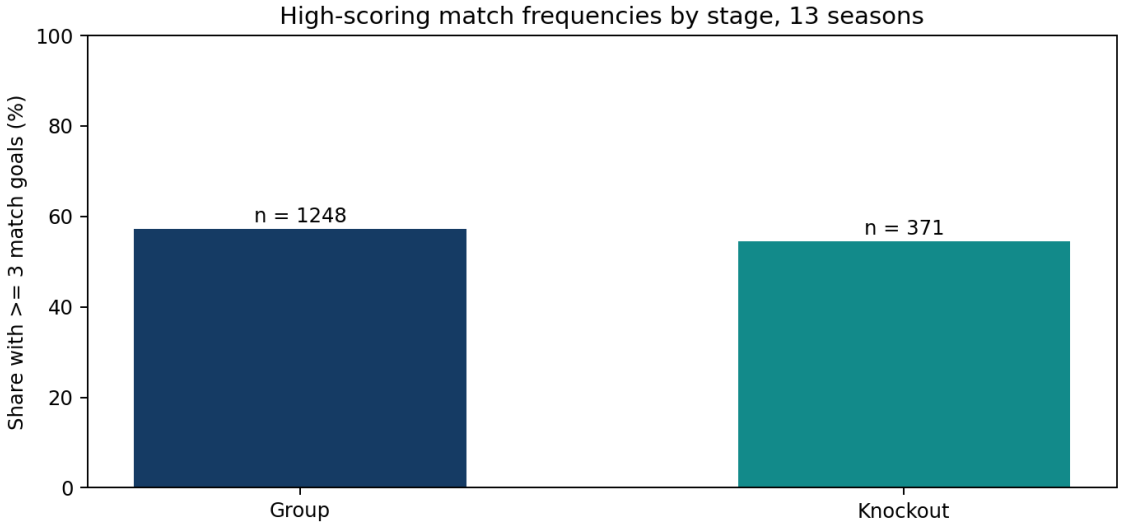


Figure 6. High-scoring proportions were 714/1,248 (57.2%) for group fixtures and 202/371 (54.4%) for knockout games.

Group-stage fixtures outnumber knockout fixtures, so the bars compare proportions. The knockout rate was 2.76 percentage points lower in this sample. Season and team strength may also influence the gap.

5.4 Seasonal Variation and Source Audit (RQ4)

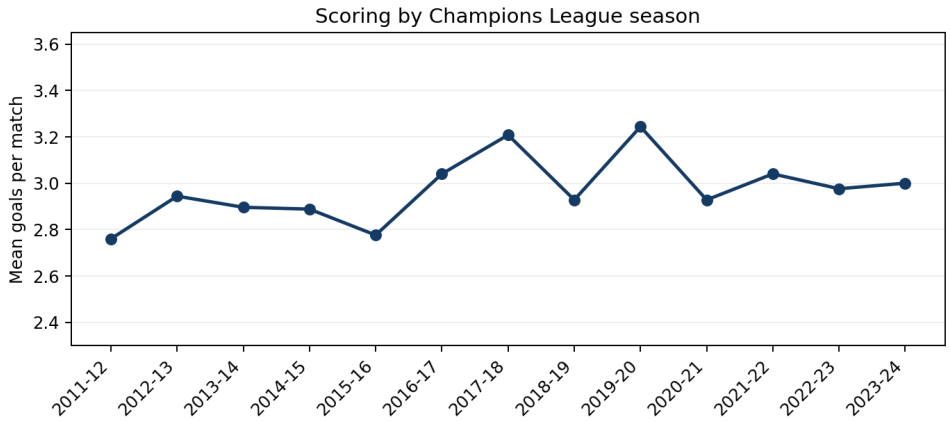


Figure 7. Seasonal average goals varied from 2.76 (2011/12) to 3.24 (2019/20), without a monotonic pattern.

The line chart highlights between-season variation in scoring opportunity and fixtures. The 2019/20 season had 119 games rather than 125, reflecting its altered knockout schedule; per-game averages remain interpretable.

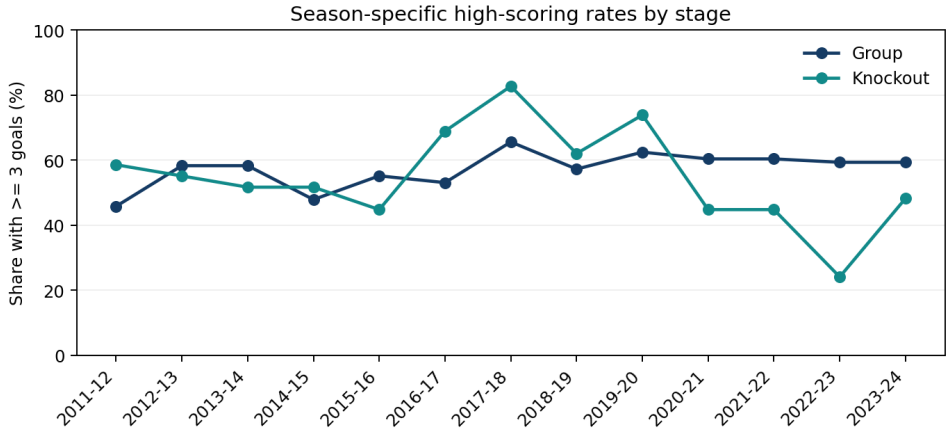


Figure 8. The high-scoring-rate gap between group and knockout games changes direction across individual seasons.

The two lines show how the scoring gap varies by season. That variation motivates the season-level uncertainty check. The plot describes an association between stage and scoring.

Pre-match predictors: Home GF and Away GF are each team's average goals scored in earlier matches that season. Home GA and Away GA are the corresponding averages for goals conceded. All four predictors are available for 1,411 fixtures.

Variable	n	Mean	SD	Min	Q1	Med	Q3	Max
Home GF	1411	1.57	0.93	0.00	1.00	1.50	2.04	6.00
Home GA	1411	1.39	0.95	0.00	0.75	1.18	2.00	7.00
Away GF	1411	1.67	0.98	0.00	1.00	1.56	2.21	7.00
Away GA	1411	1.29	0.87	0.00	0.71	1.00	1.75	6.00

6. Result Validity Challenge: Hypothesis and Assumptions

H0: Competition stage and a match's ≥3-goal indicator are independent. H1: They are associated. We used Pearson's chi-square test on the 2 × 2 table. Group matches had 534 fixtures with 0–2 goals and 714 with ≥3 goals. The corresponding knockout counts were 169 and 202.

The smallest expected cell count was 161.1, above the usual threshold of 5. The test gave $\chi^2(1) = 0.889$, $p = 0.346$, and Cramér's $V = 0.023$. The observed knockout-minus-group gap was −2.76 percentage points. A season-cluster bootstrap (3,000 draws, seed 163) gave a 95% interval of [−10.84, 4.96] percentage points, which includes zero.

Repeated clubs make the independent-match assumption questionable. The bootstrap resamples entire seasons, preserving the matches within each season as a group, but it still treats seasons as independent. We interpret the p-value and interval as exploratory.

7. Challenge Goals Update: Machine Learning

We used each team's pre-match scoring and conceding averages and one-hot encoded the stage. After excluding the 208 fixtures without complete prior-match features, we split the data by season: training on 2011/12–2019/20 (n = 975), validation on 2020/21–2021/22 (n = 218), and testing on 2022/23–2023/24 (n = 218). Regularized logistic regression and random forest were candidate models. Validation selected a random forest with maximum depth 4. We refit it on the combined training and validation sets and evaluated it on the held-out test set.

Held-out model	Accuracy	Precision	Recall	F1	ROC-AUC
Selected forest	0.546	0.542	1.000	0.703	0.473
Majority baseline	0.537	—	—	0.699	—

On the test set, the forest's F1 score (0.703) was only slightly above the majority-class baseline (0.699). Its ROC-AUC of 0.473 was close to the chance level of 0.50, suggesting that these predictors did little to distinguish high-scoring matches from other fixtures in the test seasons.

8. Work Plan Evaluation

Our proposal budgeted 44 person-hours (22 per author): 4 for setup, 8 for cleaning and parsing, 8 for historical figures, 10 for modeling, 8 for robustness checks, and 6 for final deliverables. Parsing and validation took more effort than anticipated because 2019/20 had 119 fixtures, three seasons used localized group-stage headings, and penalty shootouts needed special handling. We estimate approximately 3–4 additional person-hours for joint review, final verification, and submission preparation. We have produced the EDA figures, run the model evaluation, and completed the initial tests. The remaining work is to review the report, verify the model comparison and statistical interpretation, rerun the tests, and prepare the submission.

9. Testing and Trustworthiness

All 24 Python unittest checks pass. Tests cover seven-number summaries, missing/negative values, duplicate fixtures, expected header counts, localized stage labels, extra-time versus shootout scores, same-date no-leakage features, season resets, both bootstraps, and estimator fit/temporal partitions. We checked $12 \times 125 + 119 = 1{,}619$ games, $13 \times 96 = 1{,}248$ group games, 4,809 match goals and 916 positive target labels. All raw season files and the reproducible parser accompany this report.

10. Collaboration and Consulted Resources

Authors: Rodrigo Li and Larry Zhang. As planned, Rodrigo's focus is RQ1–RQ3 while Larry's is RQ4; both should review interpretations and tests. Sources: UEFA historical ranking pages, OpenFootball's CC0 repository, Python scientific-library documentation.

11. Limitations

UEFA ranking counts can have a different competition scope from other historical articles; two player appearance denominators remain unverified. Selected player-seasons are not a full historical census. Pre-match scoring features do not fully describe opponents, lineups or team quality; repeated teams limit simple statistical independence.

Data sources: OpenFootball Champions League (CC0) | UEFA goals | UEFA assists | UEFA appearances | UEFA season goals